Clairvoyant certificates at three quarters

A proof of Theorem 7

We prove that every window instance admits a $(\frac{3}{4}, m+3)$-certificate. The argument is the one verified in Lean, where the theorem is `threeQuarterCertificates`, proved in the module `WindowCertificates` of `ConsistentSubmodular/Deterministic/Certificates/` and its imports, and Appendix B lists the module behind each statement. Section 1 states the theorem and Section 2 reduces it to a finite coverage model. Section 3 collects basic facts and removes an initial plateau, and Sections 4–7 develop the tools: pair points and their exits, four constructors, relays to better carriers, and two rollbacks to a previous pair. Section 8 combines them. Inequalities between real numbers are collected in Appendix A, each with a certificate; every identity in this note has also been checked by exact polynomial arithmetic.

1 The statement

Let $f\colon 2^U \to \mathbb{R}_{\ge 0}$ be monotone and submodular. A *window instance* with stages $0,\dots,m$ consists of f, a set $V \subseteq U$, a budget κ, arrival sets $\emptyset = R_0 \subseteq R_1 \subseteq \cdots \subseteq R_m \subseteq U \setminus V$, and candidates $O_0,\dots,O_m \subseteq V$ with $|O_j| \le \kappa$ and $f(O \cup R_j) \le f(O_j \cup R_j)$ for all $O \subseteq V$ with $|O| \le \kappa$. Write $[m] = \{0,\dots,m\}$, $\Delta = \{p \in \mathbb{R}^{[m]}_{\ge 0} : \sum_i p_i = 1\}$ and e_i for the vertices of Δ (written δ_i in the paper). For $p \in \Delta$ let $I(p) \subseteq [m]$ contain each i independently with probability p_i, and put

$$H_j(p) = \mathbb{E}\, f\Big(R_j \cup \bigcup_{i \in I(p)} O_i\Big).$$

A *path* is a finite sequence of points of Δ joined by segments, its *pieces*; it is *safe for stage j at level ℓ* if $H_j \ge \ell$ at all of its points.

Definition 1.1 (Definition 3.1 of the paper)**.** An (α, L)-*certificate* is a point $p \in \Delta$ with paths γ^- from e_0 to p and γ^+ from p to e_m, with at most L pieces in total, such that $H_j(p) \ge \alpha f(O_j \cup R_j)$ for all j, γ^- is safe for stage 0 at level $\alpha f(O_0)$, and γ^+ is safe for stage m at level $\alpha f(O_m \cup R_m)$.

Theorem 1.2 (Theorem 7)**.** *Every window instance with stages $0,\dots,m$ admits a $(\frac{3}{4}, m+3)$-certificate. Its point has at most two nonzero coordinates.*

The second sentence is not part of Theorem 7 of the paper; the proof gives it at no extra cost. Theorem 1.2 follows from Theorem 2.2 and Proposition 2.3 below.

2 Coverage profiles

Definition 2.1 (Profiles)**.** A *profile* with stages $[m]$ is a finite family of *atoms* α, each with a *pool* $j_\alpha \in [m]$, a nonempty *label set* $L_\alpha \subseteq \{0,\dots,j_\alpha\}$ and a weight $\omega_\alpha \ge 0$, such that $m \in L_\alpha$ whenever $j_\alpha = m$ and $\omega_\alpha > 0$. For $i, t \in [m]$ and $p \in \Delta$ put

$$\lambda_t = \sum_{j_\alpha < t} \omega_\alpha, \qquad r_i(t) = \sum_{j_\alpha \ge t,\ i \in L_\alpha} \omega_\alpha, \qquad g_t = r_t(t), \qquad P_t = \lambda_t + g_t,$$

$$H_t(p) = \lambda_t + \sum_{j_\alpha \ge t} \omega_\alpha \Big(1 - \prod_{i \in L_\alpha} (1 - p_i)\Big).$$

The profile is *canonical* if $g_0 = 1$ and $r_i(t) \le g_t$ for all i, t.

Labels are candidates. An atom of pool j is covered by its labels at stages $t \le j$ and by the arrivals afterwards: λ_t is the value of the arrivals, $r_i(t)$ the residual value of candidate i, and $P_t = H_t(e_t)$ the benchmark.

Theorem 2.2 (Finite checkpoint theorem)**.** *Every canonical profile has a point $p \in \Delta$ with at most two nonzero coordinates and paths γ^- from e_0 to p and γ^+ from p to e_m, with at most $m+3$ pieces in total, such that*

$$H_t(p) \ge \tfrac{3}{4}P_t \quad (t \in [m]), \qquad H_0 \ge \tfrac{3}{4} \text{ on } \gamma^-, \qquad H_m \ge \tfrac{3}{4}P_m \text{ on } \gamma^+.$$

We call (p, γ^-, γ^+) a *certificate of the profile*, γ^- an *entry* and γ^+ an *exit*.

Proposition 2.3. *Theorem 2.2 implies Theorem 1.2, with the same point and paths.*

Proof. Let $h_t(I) = f(R_t \cup \bigcup_{i\in I} O_i)$ for $I \subseteq [m]$, so that H_t is the multilinear extension of h_t and $f(O_t \cup R_t) = h_t(\{t\})$. As f is monotone and submodular and $R_s \subseteq R_t$ for $s \le t$, each h_t is monotone and

$$h_s(I) \le h_t(I), \qquad h_t(J) - h_t(I) \le h_s(J) - h_s(I) \qquad (s \le t,\ I \subseteq J); \tag{2.1}$$

moreover $h_t(\{i\}) \le h_t(\{t\})$ by optimality of O_t. Put $\lambda_t = h_t(\emptyset)$, $r_i(t) = h_t(\{i\}) - \lambda_t$ and $g_t = r_t(t) = \max_i r_i(t)$. By (2.1) each r_i is nonincreasing, so $g_t \le r_t(0) \le g_0$. Let $\ell_{ij} = r_i(j) - r_i(j+1)$ for $j < m$ and $\ell_{im} = r_i(m)$, and let a maximum over the empty set be 0, as $I(p)$ may be empty. For $i \in I$ and $j < m$, (2.1) applied to $\{i\} \subseteq I$ gives $[h_j(I) - \lambda_j] - [h_{j+1}(I) - \lambda_{j+1}] \ge \ell_{ij}$; summing over $j \ge t$ and using $h_m(I) - \lambda_m \ge \max_{i\in I} r_i(m)$, we get for every $I \subseteq [m]$ (all terms vanish if $I = \emptyset$)

$$h_t(I) - \lambda_t \ \ge\ \sum_{j\ge t} \max_{i\in I} \ell_{ij} \ \ge\ \sum_{j\ge t} \max_{i\in I} \hat\ell_{ij}, \qquad \hat\ell_{ij} := \ell_{ij} \cdot [\, i \le j \,]. \tag{2.2}$$

The right side is a coverage function: cut the interval $[0, \max_i \hat\ell_{ij})$ of pool j at the points $\hat\ell_{ij}$; each piece $[y, y')$ becomes an atom of pool j with weight $y' - y$ and labels $\{i : \hat\ell_{ij} \ge y'\} \subseteq \{0, \dots, j\}$. In this profile $\hat r_i(t) = r_i(\max\{i,t\}) \le g_t$ and $\hat g_t = g_t$, and label m covers pool m since $\hat\ell_{mm} = g_m = \max_i \hat\ell_{im}$. By (2.1), $\lambda_{j+1} - \lambda_j \ge \ell_{ij}$ for all i, so $\sigma_t := \lambda_t - \hat\lambda_t \ge \lambda_0 \ge 0$. Taking expectations in (2.2), $H_t(p) \ge \hat H_t(p) + \sigma_t$, while $f(O_t \cup R_t) = \hat P_t + \sigma_t$. Hence $H_t(p) - \frac{3}{4} f(O_t \cup R_t) \ge \hat H_t(p) - \frac{3}{4}\hat P_t$ for all p and t, and a certificate of the new profile is a $(\frac{3}{4}, m+3)$-certificate of the window. If $g_0 > 0$, divide all weights by g_0 to make the profile canonical. If $g_0 = 0$, all $\hat r_i$ vanish, $\hat H_t \equiv \hat P_t$, and e_0 with the exit $[e_0, e_m]$ is a certificate. □

3 Basic facts and the initial plateau

Lemma 3.1. *In a canonical profile:*

- *(a) each r_i is nonincreasing, $r_i(t) = g_i$ for $t \le i$, and $1 = g_0 \ge g_1 \ge \cdots \ge g_m \ge 0$;*
- *(b) (persistence) $\lambda_t + r_i(t) \ge \lambda_s + r_i(s)$ for $s \le t$; hence $1 \le P_s \le P_t$;*
- *(c) for $a \ne c$ and $x \in [0,1]$, $H_t((1-x)e_a + xe_c) = \lambda_t + (1-x)r_a(t) + x\, r_c(t) - x(1-x)\, o_{ac}(t)$, where $o_{ac}(t) = \sum_{j_\alpha \ge t,\ a,c\in L_\alpha} \omega_\alpha$ satisfies $o_{ac}(t) \le \min\{r_a(t), r_c(t)\}$, and $o_{ac}(t) \le r_a(c)$ if $a < c$; also $H_t(e_i) = \lambda_t + r_i(t)$;*
- *(d) if $\lambda_t \ge 3g_t$ (stage t is* inactive*), then $H_t \ge \frac{3}{4}P_t$ on all of Δ;*
- *(e) if $p \in \Delta$ is supported on $\{t, \dots, m\}$, then $H_t(p) - \frac{3}{4}P_t \ge H_0(p) - \frac{3}{4}$.*

Proof. Atoms with label i lie in pools $\ge i$; this gives (a), as $g_i = r_i(t) \le g_t$ for $t \le i$, and the bound $o_{ac} \le r_a(c)$. In (b), an atom of a pool in $[s,t)$ leaves r_i and enters λ; take $i = s$ and use $r_s(t) \le g_t$. In (c), an atom labelled by both a and c is missed with probability $x(1-x)$. (d) holds as $H_t \ge \lambda_t$. In (e), the atoms meeting the support of p lie in pools $\ge t$, so $H_t(p) = \lambda_t + H_0(p)$, while $P_t \le \lambda_t + 1$. □

Lemma 3.2 (Initial plateau)**.** *If Theorem 2.2 holds for* strict *profiles, i.e. canonical profiles with $g_t < 1$ for all $t \ge 1$, then it holds for all canonical profiles.*

Proof. Let h be the last stage with $g_h = 1$. If $h = 0$ the profile is strict. If $h = m$, then $H_t(e_m) = \lambda_t + 1 \geq P_t$ for all t, and the edge $[e_0, e_m]$ has $H_0 \geq 1 - x(1-x) \geq \frac{3}{4}$ by Lemma 3.1(c), as $r_0(0) = r_m(0) = 1$; so e_m with entry $[e_0, e_m]$ is a certificate. Otherwise form the suffix profile with stages $h, \ldots, m$: keep the atoms of pools $\geq h$, replace L_α by $L_\alpha \cap \{h, \ldots, m\}$, and delete the atoms for which this is empty. Residuals of labels $\geq h$ are unchanged, so the suffix profile is strict. Let $c_t \geq 0$ be λ_h plus the weight deleted from pools $h, \ldots, t-1$. Writing tildes for the suffix profile, every p supported on $\{h, \ldots, m\}$ satisfies

$$H_t(p) = \tilde{H}_t(p) + c_t, \quad P_t = \tilde{P}_t + c_t \quad (t \geq h), \qquad H_t(p) = \lambda_t + \tilde{H}_h(p), \quad P_t = \lambda_t + 1 \quad (t < h).$$

Hence a certificate of the suffix profile satisfies all stage inequalities, its exit is an exit, and its entry, preceded by the edge $[e_0, e_h]$ (safe as for $h = m$), is an entry. The suffix has $m - h + 1$ stages, so there are at most $(m-h) + 3 + 1 \leq m + 3$ pieces. □

From now on the profile is strict.

4 Carriers, pair points and exits

Definition 4.1 (Carriers)**.** A *carrier* is a type a with $A := g_a \geq \frac{3}{4}$. Put

$$d = 1 - A, \qquad D = 3d, \qquad B = 4A - 3 = A - D, \qquad V_a = P_a - 1,$$

so $\lambda_a = V_a + d$, and for $t \geq a$ put

$$\delta_t = A - g_t, \qquad w_t = g_t - r_a(t), \qquad v_t = P_t - A.$$

By Lemma 3.1, $\delta_t, w_t \geq 0$, $\delta_t + w_t = A - r_a(t) \leq A$, δ_t and v_t are nondecreasing, and persistence of a gives $v_t \geq \lambda_a + w_t$. The carrier has *strong credit* if $V_a \geq \frac{1}{2}$ or $d \leq V_a(1 - V_a)$; this implies $V_a \geq d$ and $\lambda_a \geq 2d$. The root $a = 0$ has $A = 1$ and $d = V_0 = \lambda_0 = 0$, so it has strong credit.

For active $t > a$ with $\delta_t + w_t > 0$ let $\rho_t = (4w_t - v_t - D)/(\delta_t + w_t)$. The *envelope* q_a is the maximum of 0 and these ratios, and an *attainer* is an active $t > a$ with $\delta_t + w_t > 0$ and $\rho_t = q_a$. For $q \geq q_a$ and all active $t > a$,

$$K_t(q) := q(\delta_t + w_t) + v_t + D - 4w_t \;\geq\; 0$$

(if $\delta_t + w_t = 0$ then $w_t = 0$). Also $q_a < 3$, since $4w_t - v_t - D \leq 3w_t - D$, and either $D > 0$ or $a = 0$ and $\delta_t > 0$.

For $0 < q < 3$ put $x = q/4$, $\mu = 4x(1-x) = q - q^2/4 \in (0, 1]$, $\theta = \theta_a(q) = 2A - 3B/q$, and for $c > a$

$$E_c = B - q\,\delta_c - \mu\, r_a(c), \qquad S_c = w_c - (q-1)\delta_c - D, \qquad p_c = (1-x)e_a + xe_c.$$

A row c is *actual-eligible* if $E_c \geq 0$ and *sufficient-eligible* if $S_c \geq 0$. Since $E_c - S_c = (1-\mu)r_a(c) \geq 0$, sufficient-eligible rows are actual-eligible. An *access path* of a is a path from e_0 to e_a with at most a pieces that is safe for stage 0 at level $\frac{3}{4}$.

Lemma 4.2 (Pair rows)**.** *Let a be a carrier, $0 < q < 3$ with $q \geq q_a$, $c > a$ and $p = p_c$.*

(a) $H_t(p) - \frac{3}{4}P_t \geq \frac{1}{4}E_c$ for $t \leq a$, and $4H_t(p) - 3P_t \geq E_c + K_t(q)$ for active $t \in (a, c]$. Both bounds also hold with S_c in place of E_c.

(b) On the segment $[e_a, p]$, $H_0 \geq \frac{3}{4} + \frac{1}{4}S_c$; if $q \leq 2$, also $H_0 \geq \frac{3}{4} + \frac{1}{4}E_c$.

(c) For $t > c$, with $u = r_a(t)$ and $y = r_c(t)$, and for $\nu \in \{\mu, 1\}$,

$$4H_t(p) - 3P_t \;\geq\; A + v_c + (4-q)u + (q-1)y - \nu \min\{u, y\} - 3g_t.$$

Proof. For $t \le c$, Lemma 3.1(a),(c) give $r_c(t) = g_c$ and $o_{ac}(t) \le r_a(c)$, so for $\nu \in \{\mu, 1\}$

$$4H_t(p) - 3P_t \ge \lambda_t + (4-q)r_a(t) + q\,g_c - \nu\,r_a(c) - 3g_t.$$

For active $t \in (a, c]$, substituting $\lambda_t = v_t + \delta_t$, $r_a(t) = A - \delta_t - w_t$, $g_t = A - \delta_t$ and $g_c = A - \delta_c$ turns the right side into exactly $E_c + K_t(q)$ for $\nu = \mu$, and $S_c + K_t(q)$ for $\nu = 1$. For $t \le a$, Lemma 3.1(e) reduces to $t = 0$, where $r_a(0) = A$ and $H_0(p) \ge A - x\delta_c - x(1-x)r_a(c) = \frac{3}{4} + \frac{1}{4}E_c$, and also $H_0(p) \ge A - x\delta_c - \frac{1}{4}r_a(c) = \frac{3}{4} + \frac{1}{4}S_c$. For (b), these two bounds with x replaced by $s \in [0, x]$ are nonincreasing in s (the first because $s \le x \le \frac{1}{2}$). For (c), use $o_{ac}(t) \le \min\{u, y\}$, $4x(1-x) = \mu \le 1$ and persistence of c, which gives $\lambda_t \ge P_c - y = A + v_c - y$. □

Every atom of pool m contains the label m. Hence, for each type $i < m$ and $y \in [0, 1]$,

$$H_m((1-y)e_i + ye_m) = \lambda_m + (1-y)^2 r_i(m) + y\,g_m. \tag{4.1}$$

This is convex in y. For $i = m$ the segment is the single point e_m, with value P_m. If $i < m$ and e_i is safe, i.e. $\lambda_m + r_i(m) \ge \frac{3}{4}P_m$, then the edge $[e_i, e_m]$ is safe for stage m at level $\frac{3}{4}P_m$ unless its minimum is interior and too low. That happens only if

$$r_i(m) > \tfrac{1}{2}g_m, \qquad r_i(m)(\lambda_m + g_m) < g_m^2, \qquad \text{hence} \quad \lambda_m < g_m. \tag{4.2}$$

Persistence gives $\lambda_m + r_i(m) \ge P_i \ge g_i \ge g_m$ for every i.

Lemma 4.3 (Exits)**.** *Let $a < c$, $x \in [0, 1]$ and $p = (1-x)e_a + xe_c$ with $H_m(p) \ge \frac{3}{4}P_m$. Suppose one of the following:*

(a) $\lambda_m + (1-x)r_a(m) + x\,r_c(m) - \frac{1}{4}\min\{r_a(m), r_c(m)\} \ge \frac{3}{4}P_m$;

(b) a is a carrier, $x = q/4$ with $q_a \le q \le 2$, and $q \ge 1$ or $\lambda_a \ge \frac{3}{2}d$.

Then for some $b \in \{a, c\}$ the path $p \to e_b \to e_m$ is safe for stage m at level $\frac{3}{4}P_m$.

Proof. Write $L = \lambda_m$, $G = g_m$ and $C = \frac{3}{4}(L + G)$. If m is inactive, all of Δ is safe at stage m; so assume m is active.

(a) Let $b \in \{a, c\}$ maximize $R := r_b(m)$, and let $w = \min\{r_a(m), r_c(m)\}$. Along $[p, e_b]$ the weighted average of $r_a(m)$ and $r_c(m)$ does not decrease. Moreover H_m dominates $L + (\text{average}) - w/4$ along it, since $\frac{w}{4} - z(1-z)o = w(z - \frac{1}{2})^2 + z(1-z)(w - o) \ge 0$. So $[p, e_b]$ is safe and $L + R - \frac{w}{4} \ge C$. If $[e_b, e_m]$ were unsafe, (4.2) would give $L < G$; but then $w \ge G - L$ yields $R \ge \frac{1}{4}(3G - L + w) \ge G - \frac{L}{2}$, and $(G - \frac{L}{2})(G + L) - G^2 = \frac{1}{2}L(G - L) \ge 0$ contradicts $R(L + G) < G^2$.

(b) Let $u = r_a(m)$ and $p' = (1-x)e_a + xe_m$. Comparing atom by atom (every atom of pool m contains m), $H_m(p') \ge H_m(p)$. First, the segment $[p', e_m]$ is safe. By (4.1) with $i = a < m$ it is unsafe only if the minimizer of $s \mapsto L + (1-s)^2 u + sG$ lies in $(x, 1)$ with value below C. That means $u(L+G) < G^2$ and $2u(1-x) > G$. For $q = 2$ the second fails, as $u \le G$. For $q < 2$, Fact A.2 excludes both. It applies with $\lambda = \lambda_a$ (note $A + \lambda_a = P_a \ge 1$), because persistence gives $L + u \ge A + \lambda_a$, and $K_m(q) \ge 0$ reads $q(A - u) \ge 3G - 4u - L + 4A - 3$.

Second, if $p'' \in \Delta$ has $p''_i = z$, then $H_m(p'') \le L + G(1-z) + r_i(m)z^2$. Indeed, an atom containing i is missed with probability at least $(1-z)(1-\sum_{j \ne i} p''_j) \ge z(1-z)$, and every other atom with probability at least z. Hence, if e_i is safe but $[e_i, e_m]$ is not, every p'' with $p''_i = z_i := G/(2r_i(m)) \in (\frac{1}{2}, 1)$ is unsafe, by (4.2).

Third, H_m is a convex quadratic along the line through e_a and e_c (Lemma 3.1(c)). So one of $[p, e_a]$ and $[p, e_c]$, say $[p, e_b]$, is safe. If $b = m$ we are done. Suppose $[e_b, e_m]$ is unsafe. The safe segment $[p, e_b]$ cannot meet the slice $\{p''_b = z_b\}$, so $p_b > z_b > \frac{1}{2}$. Since $p_c = x \le \frac{1}{2}$, we have $b = a$ and $1 - x > z_a$. But then the safe segment $[p', e_m]$, whose a-coordinate decreases from $1 - x$ to 0, meets the slice $\{p''_a = z_a\}$, a contradiction. □

5 Constructors

Lemma 5.1 (Constructors)**.** *Let a be a carrier with an access path π, and let p be as in one of the cases below (eligibility and θ refer to the stated q). Then $H_t(p) \geq \frac{3}{4}P_t$ for all t. The path π followed by $[e_a, p]$ is an entry, and p has an exit of at most two pieces. These form a certificate with at most $a+3$ pieces.*

- *(V)* $q_a = 0$*, a has strong credit, and $p = e_a$.*
- *(L)* $0 < q := q_a \leq 1$*, a has strong credit, some active $c > a$ is actual-eligible, and $p = p_c$ for the last such c.*
- *(A)* $1 \leq q \leq 2$*, $q \geq q_a$, some active $c > a$ is actual-eligible, $p = p_c$ for the last such c, and $v_c \geq \theta$.*
- *(S)* $1 \leq q < 3$*, $q \geq q_a$, some active $c > a$ is sufficient-eligible, $p = p_c$ for the last such c, and $v_c \geq \max\{2D, \theta\}$.*

Proof. Inactive stages are harmless by Lemma 3.1(d). In (L), (A) and (S), the stages $t \leq c$ and the entry follow from Lemma 4.2(a),(b), because $E_c \geq 0$ (resp. $S_c \geq 0$), $K_t(q) \geq 0$, and $q \leq 2$ in (L) and (A). Let $t > c$ be active. By maximality of c, $E_t < 0$ (resp. $S_t < 0$). Write $\delta = \delta_t$ and $w = w_t$, so that $u = A - \delta - w$ and $g_t = A - \delta$ in Lemma 4.2(c).

(A) Take $\nu = \mu$. The bound depends on y through $(q-1)y - \mu\min\{u, y\}$, with slope $q^2/4 - 1 \leq 0$ for $y < u$ and $q - 1 \geq 0$ for $y > u$. At $y = u$ it equals $v_c - F$, where $F = (3-\mu)w - \mu\delta - (1-\mu)A$. Since $E_t < 0$ reads $\mu w < \mu A - B + (q-\mu)\delta$, Fact A.1 gives $F \leq \theta \leq v_c$.

(L) Take $\nu = \mu$. As $q \leq 1$, the y-dependent part is nonincreasing and $y \leq g_t$. At $y = g_t$ the bound equals $(1-\mu)A + v_c + \mu\delta - (4-q-\mu)w$. Multiply $\mu w < \mu A - B + (q-\mu)\delta$ by $(4-q-\mu)/\mu$ and use $(4-q-\mu)q = (4-q)\mu$ and $(4-q-\mu)(q-\mu) = \mu^2$. This gives $(4-q-\mu)w - \mu\delta < (\frac{4}{q} - 1)(\mu A - B)$, so the margin exceeds $v_c - (3-q)A + (\frac{4}{q} - 1)B$. On the other hand, $K_c(q) \geq 0$ and $E_c \geq 0$ give $v_c + D \geq (4-q)w_c - q\delta_c = \frac{4}{q}(E_c + \mu A - B) \geq \frac{4}{q}(\mu A - B)$, which is $v_c \geq (3-q)A - (\frac{4}{q} - 1)B$.

(S) Take $\nu = 1$. Considering $y \geq u$ and $y \leq u$ separately, the bound is at least $v_c + \delta - 2w$ if $q \leq 2$, and at least $(2-q)A + v_c + (q-1)\delta - (4-q)w$ if $q \geq 2$. Here $S_t < 0$ reads $w < D + (q-1)\delta$, and $w \leq A - \delta$.

- For $\delta \geq B/q$, $w \leq A - \delta$ bounds both subtracted quantities by $2A - 3\delta \leq \theta$.
- For $\delta \leq B/q$, $w < D + (q-1)\delta$ bounds them by $2D + (2q-3)\delta$ and $(q-2)A + (4-q)D + (q-1)(3-q)\delta$. Both are linear in δ and equal θ at $\delta = B/q$; the first equals $2D$ at $\delta = 0$, and the second is nondecreasing in δ as $q < 3$.

So the margin is at least $v_c - \max\{2D, \theta\} \geq 0$.

(V) For active $t > a$, $4H_t(e_a) - 3P_t = A + v_t - 4w_t = B + K_t(0) \geq 0$. For $t \leq a$, use Lemma 3.1(e) and $H_0(e_a) = A$. The entry is π itself.

Exits. In (V) and (L) strong credit gives $\lambda_a \geq 2d$, and Lemma 4.3(b) applies; in (V) take $x = 0$ and $c = m$, and if $a = m$ there is nothing to do. In (A), Lemma 4.3(b) applies since $q \geq 1$. In (S), the terminal row was proved through the $\nu = 1$ bounds: by Lemma 4.2(c) if $m > c$, and by Lemma 4.2(a) if $m = c$, where $r_a(c) = \min\{r_a(m), r_c(m)\}$ because $r_c(m) = g_m$. Hence Lemma 4.3(a) applies. The piece count is $|\pi| + 1 + 2 \leq a + 3$. □

6 Relays

For a carrier a and $s > a$ let $J_{as}(z) = A - (A - g_s)z - r_a(s)z(1-z) - \frac{3}{4}$ for $z \in [0,1]$. By Lemma 3.1(a),(c), $H_0((1-z)e_a + ze_s) \geq \frac{3}{4} + J_{as}(z)$. The edge $[e_a, e_s]$ is *safe* if $J_{as} \geq 0$ on $[0,1]$, and *strictly safe* if $J_{as} > 0$ there. A safe edge extends an access path of a to one of s, and $J_{as}(1) \geq 0$ means $g_s \geq \frac{3}{4}$.

Lemma 6.1 (Credit transfer)**.** *If a is a carrier with $A > \frac{3}{4}$ and $[e_a, e_s]$ is safe, then s has strong credit.*

Proof. Persistence gives $V_s = P_s - 1 \geq V_a + w_s$. If $V_s \geq \frac{1}{2}$ we are done. Otherwise $V_a + w_s < \frac{1}{2}$, and Fact A.3 (with $v = V_a$, $e = \delta_s$, $w = w_s$) gives $1 - g_s = d + \delta_s \leq (V_a + w_s)(1 - V_a - w_s) \leq V_s(1 - V_s)$. □

Lemma 6.2 (Attainers)**.** *Let a be a carrier with strong credit and $q = q_a > 0$, let s be an attainer, and let $b = v_s - w_s$. Then $b \geq \lambda_a$ and $(3-q)w_s = q\delta_s + b + D$. Moreover:*

(a) $(3-q)S_s = b - (2-q)D + (q^2 - 3q + 3)\delta_s$; so s is sufficient-eligible if $q \geq \frac{4}{3}$, or if $q \geq 1$ and $\lambda_a \geq D$;

(b) if $1 \leq q \leq 2$ and $A \geq \frac{33}{40}$, then s is actual-eligible (Fact A.4);

(c) if $0 < q \leq 1$ and $A \geq \frac{1281}{1600}$, then s is actual-eligible (Fact A.5);

(d) if $1 \leq q \leq \frac{3}{2}$ and $\lambda_a \geq D$, then $v_s \geq \theta(q)$ (Fact A.6).

Proof. The first claim is persistence and the second is $\rho_s = q$. In (a), $b - (2-q)D \geq 0$ because $b \geq \lambda_a \geq 2d = \frac{2}{3}D$ in the first case, and $b \geq D$ in the second. In (b), Fact A.4 applies if $d > 0$; if $d = 0$, the identity for $4E$ in Appendix A reads $4E_s = qv_s + (q-2)^2 \geq 0$. □

Lemma 6.3 (Strict relays)**.** *Let a be a carrier with strong credit and $A > \frac{3}{4}$, let $1 \leq q = q_a < 3$, and let s be an attainer with $v_s < \theta(q)$. If $\lambda_a \geq D$, or $q \geq \frac{4}{3}$, or $A \geq \frac{33}{40}$, then $[e_a, e_s]$ is strictly safe. In particular $g_s > \frac{3}{4}$.*

Proof. Write $\delta = \delta_s$, $w = w_s$, $b = v_s - w_s$, $r = r_a(s) = A - \delta - w$, $M = (A-w)(w-D) - B\delta$ and $\mathcal{E}(\beta, \omega) = \beta^2 - \beta(2A + 2\omega - D) + D(A + \omega)$. Since $q(\delta + w) = 3w - b - D$, expanding gives

$$(\delta + w)\big[(2A - v_s)q - 3B\big] = 3M + \mathcal{E}(b, w), \tag{6.1}$$

and the left side is positive. As $J_{as}(z) = \frac{B}{4} - (\delta + r)z + rz^2$ and $(\delta + r)^2 - Br = -M$, the condition $M > 0$ forces $r > 0$ and leaves J_{as} with no real root, so the edge is strictly safe. It therefore suffices to show $\mathcal{E}(b, w) \leq 0$.

Here $b \geq \lambda_a$, $b < 2A - w$ (as $v_s < \theta \leq 2A$) and $w \leq A$. Also $\mathcal{E}$ is convex in β, with $\mathcal{E}(D, w) = D(2D - A - w) \leq 0$ and $\mathcal{E}(2A - w, w) = 3(AD - 2Aw + w^2) \leq 0$ whenever $D \leq w \leq A$.

- If $\lambda_a \geq D$, then $(3-q)w \geq b + D \geq 2D$ gives $w \geq D$, and convexity on $[D, 2A - w]$ gives $\mathcal{E}(b, w) \leq 0$.
- Otherwise $\lambda_a < D$, so $d > 0$, $V_a < 2d \leq \frac{1}{2}$ and $d \leq V_a(1 - V_a)$. Moreover $\mathcal{E}(\lambda_a, \cdot)$ is decreasing, with slope $D - 2\lambda_a < 0$.
  - If $q \geq \frac{3}{2}$, then $w \geq \frac{\lambda_a + D}{3-q} \geq \frac{2}{3}(\lambda_a + D) > D$, and $\mathcal{E}(\lambda_a, w) \leq \mathcal{E}(\lambda_a, \frac{2}{3}(\lambda_a + D)) < 0$ by Fact A.7.
  - If $A \geq \frac{33}{40}$, Fact A.8 gives $w > D$, so $\mathcal{E}(\lambda_a, w) < \mathcal{E}(\lambda_a, D) < 0$.

  In both cases convexity on $[\lambda_a, 2A - w]$ gives $\mathcal{E}(b, w) \leq 0$.

The remaining case, $\lambda_a < D$, $A < \frac{33}{40}$ and $\frac{4}{3} \leq q < \frac{3}{2}$, is Fact A.9, which proves $J_{as} > 0$ directly. □

Lemma 6.4 (Inheritance)**.** *Let a be a carrier with strong credit and $A > \frac{3}{4}$, let $k = q_a \in [1, 3)$, and let h be an attainer with $g_h \geq \frac{3}{4}$ and $v_h < \theta_a(k)$.*

(a) If $k < \frac{4}{3}$ and $A \geq \frac{33}{40}$, then $g_h > \frac{1281}{1600}$ (Fact A.10).

(b) If $\lambda_h < 3(1 - g_h)$, then $\frac{4}{3} < k < 2$ and $1 - A < \frac{1}{8}$ (Fact A.11).

Proof. Both parts apply the cited facts with $e = \delta_h$, $W = w_h$ and $b = v_h - w_h$ (Lemma 6.2). Fact A.10 needs $A < 1$, which holds in (a): $A = 1$ would give $\theta_a(k) = 2 - \frac{3}{k} < 0 \leq v_h$. □

7 Rollbacks

Definition 7.1 (Relayed carriers)**.** A *relayed carrier* is a carrier h with $\frac{3}{4} < g_h < 1$, strong credit and an access path, together with a *parent* $a < h$ and $k := q_a$ such that

(H1) a is a carrier with $g_a > \frac{3}{4}$, strong credit and an access path π_a, and $1 \leq k < 3$;

(H2) h is an attainer of a, and $[e_a, e_h]$ is safe;

(H3) the *parent test failed*: $v_t^a < \theta_a(k)$ for every active $t > a$ that is actual-eligible (if $k \leq 2$), resp. sufficient-eligible (if $k > 2$), for a at k;

(H4) if $k < \frac{4}{3}$ then $g_a \geq \frac{33}{40}$.

Superscripts a refer to the notation of Definition 4.1 for a. Condition (H3) holds at every sufficient-eligible row, and in particular at h: h is sufficient-eligible if $k \geq \frac{4}{3}$ and actual-eligible otherwise, by Lemma 6.2(a),(b) and (H4). We write

$$d_0 = 1 - g_a, \qquad d = 1 - g_h, \qquad p = P_h - 1, \qquad W = g_h - r_a(h),$$
$$C = 2 - \frac{3}{k} + \frac{3(4-k)d_0}{k}, \qquad \hat{x} = \Big(1 - \frac{k}{4}\Big)e_a + \frac{k}{4}e_h.$$

Lemma 7.2 (Rows of the rollback point)**.** *Let h be a relayed carrier with $k \geq \frac{4}{3}$.*

(a) $p = (4-k)W - kd - (4-k)d_0$, $p \geq W + V_a$ and $p < C$. An active $t > a$ is sufficient-eligible for a if and only if $g_t - r_a(t) \geq (k-1)(1-g_t) + (4-k)d_0$, and then $P_t - 1 < C$.

(b) $H_t(\hat{x}) \geq \frac{3}{4}P_t$ for $t \leq h$, and π_a followed by $[e_a, \hat{x}]$ is an entry.

(c) For active $t > h$ let $T = 1 - g_t$, $P' = P_t - 1$, $u = g_t - r_h(t)$, $z = g_t - r_a(t)$ and $\mathrm{cap} = (k-1)T + (4-k)d_0$. Then

$$4H_t(\hat{x}) - 3P_t \;\geq\; M := T + P' - (4-k)z - ku + \max\{z, u\},$$

the sufficient row with overlap $\min\{r_a(t), r_h(t)\}$. Moreover $P' \geq p + u$, $P' \geq p + z - W$ and $u \leq 1 - T$.

(d) In (c): if t is sufficient-eligible for a, then $M \geq U := d - 2C + 3p - (3-k)W$. If not, then $M \geq p + (3-2k)T - 2(4-k)d_0$ when $k \leq 2$, and $M \geq p - (k-2) - (k-1)(3-k)T - (4-k)^2 d_0$ when $k \geq 2$.

(e) If $M \geq 0$ at $t = m$ whenever $m > h$ is active, then $\hat{x}$ has an exit of at most two pieces.

Proof. (a) In child notation, $\delta_h^a = d - d_0$, $w_h^a = W$, $v_h^a = p + d_0$ and $D^a = 3d_0$. So $\rho_h^a = k$ is the first identity. Persistence of a gives $p + d_0 \geq \lambda_a + W = V_a + d_0 + W$. Next, $\theta_a(k) + g_a - 1 = C$, so $v_t^a < \theta_a(k)$ reads $P_t - 1 < C$, and (H3) at h gives $p < C$. Finally $S_t^a(k) = z - (k-1)(1 - g_t - d_0) - 3d_0$.

(b) h is sufficient-eligible for a (Lemma 6.2(a)). So Lemma 4.2(a), with $c = h$, $q = k$ and S in place of E, proves the rows $t \leq h$. The segment $[e_a, \hat{x}]$ lies in the safe edge $[e_a, e_h]$.

(c) Use Lemma 3.1(c) with $4x(1-x) \leq 1$, $o_{ah}(t) \leq \min\{r_a(t), r_h(t)\}$ and $\lambda_t = P' + T$. The three inequalities are persistence of h and of a from h to t, and $r_h(t) \geq 0$.

(d) M equals $M_1 := T + P' - (3-k)z - ku$ if $z \geq u$, and $M_2 := T + P' - (4-k)z - (k-1)u$ if $z \leq u$; always $M \geq M_2$. If t is eligible, then $P' < C$, and

$$M_1 - U = (T - d) + 2(C - P') + (3-k)(P' - p - z + W) + k(P' - p - u),$$
$$M_2 - U = (T - d) + 2(C - P') + (3-k)W + 3(P' - p - u) + (4-k)(u - z).$$

If t is not eligible, then $z < \mathrm{cap}$. For $k \leq 2$,

$$M_1 - (T + p - 2z) = (P' - p - u) + (k-1)(z - u), \qquad M_2 - (T + p - 2z) = (P' - p - u) + (2-k)(u - z),$$

and $T + p - 2z \geq T + p - 2\,\mathrm{cap}$. For $k \geq 2$,

$$M_2 - \big[p - (k-2) - (k-1)(3-k)T - (4-k)^2 d_0\big] = (P' - p - u) + (k-2)(1 - T - u) + (4-k)(\mathrm{cap} - z).$$

(e) If $m > h$, M is four times the sufficient row of Lemma 4.3(a) minus $3P_m$. If $m = h$, the row proved in (b) is that row, since $r_a(h) = \min\{r_a(m), r_h(m)\}$. Apply Lemma 4.3(a). □

Lemma 7.3 (Low rollback)**.** *Let h be a relayed carrier with $k \geq \frac{4}{3}$, $g_h < \frac{1281}{1600}$ and $0 < q := q_h \leq 1$, and suppose that no active $t > h$ is actual-eligible for h at q. Then $\hat{x}$, with the entry of Lemma 7.2(b) and the exit of Lemma 7.2(e), is a certificate with at most $a + 3$ pieces.*

Proof. Let s be an attainer of h, and put $\delta = g_h - g_s$, $w = g_s - r_h(s)$ and $P = P_s - 1$. In the notation of h, the relations $\rho_s = q$, persistence, $r_h(s) \ge 0$ and $E_s^h < 0$ read

$$P = (4-q)w - q\delta - 4d, \quad P \ge p + w, \quad w \le 1 - d - \delta, \quad qP \le (4-q)^2 d - (q-2)^2.$$

The last uses the identity $4E_s^h = qP + (q-2)^2 - (4-q)^2 d$, valid at attainers. These relations, together with Lemma 7.2(a), strong credit of a, $d_0 \le d$ and $\frac{319}{1600} < d \le \frac{1}{4}$, are the hypotheses of Fact A.12. It yields $\frac{40}{23} < k < \frac{40}{17}$, $P > C$, $U > 0$, $Y > 0$, and $V > 0$ if $k \le 2$, resp. $V' > 0$ if $k \ge 2$, where

$$\begin{aligned} V &= p + (3-2k)(d+\delta) - 2(4-k)d_0, \\ V' &= p - (k-2) - (k-1)(3-k)(d+\delta) - (4-k)^2 d_0, \\ Y &= \frac{(k-1)(3-k)}{k}\big[P - (k-1)(d+\delta) - (4-k)d_0\big] + \frac{k^2-3k+3}{k}p - C. \end{aligned}$$

By Lemma 7.2 it remains to show $M > 0$ for active $t > h$. Since P_t is nondecreasing and $P_t - 1 < C$ at eligible rows, $P > C$ puts every sufficient-eligible row before s. In particular s is not eligible: $z_s := g_s - r_a(s) \le (k-1)(d+\delta) + (4-k)d_0$.

- *Eligible t:* $M \ge U > 0$.
- *Ineligible $t \le s$:* here $T \le d + \delta$. For $k \le 2$, as $3 - 2k < 0$, we get $M \ge V > 0$. For $k \ge 2$, the bound in Lemma 7.2(d) decreases in T, so $M \ge V' > 0$.
- *Ineligible $t \ge s$:* persistence of a from s to t gives $X := P' - P - z + z_s \ge 0$, and

$$\begin{aligned} kM_2 = 3(\mathrm{cap} - z) + (k^2-3k+3)(P' - p - u) + (2k-3)(1 - T - u) \\ + (k-1)(3-k)\big[X + (k-1)(d+\delta) + (4-k)d_0 - z_s\big] + kY, \end{aligned}$$

  which is positive. □

Lemma 7.4 (Middle rollback)**.** *Let h be a relayed carrier with $g_h < \frac{33}{40}$, $\lambda_h < 3(1-g_h)$ and $1 \le q := q_h < \frac{4}{3}$, and suppose $v_t^h < \theta_h(q)$ for every active $t > h$ that is actual-eligible for h at q. Then $k > \frac{4}{3}$, and the conclusion of Lemma 7.3 holds.*

Proof. By (H3) at h and Lemma 6.4(b), $\frac{4}{3} < k < 2$ and $d_0 < \frac{1}{8}$, so Lemma 7.2 applies. Also $\frac{7}{40} < d \le \frac{1}{4}$ and $p = \lambda_h - d < 2d$. The safe edge $[e_a, e_h]$ at $z = \frac{1}{2}$ gives $W \ge d + 2d_0$. Let s, δ, w and P be as before. Then $v_s^h < \theta_h(q)$: by hypothesis if s is actual-eligible, and otherwise by the identity $4E_s^h = q(v_s^h - \theta_h(q)) + (q-1)[(q-1)g_h + 3d]$, valid at attainers. So

$$P = (4-q)w - q\delta - 4d, \quad P \ge p + w, \quad w \le 1 - d - \delta, \quad q(2 - 3d - P) \ge 3(1 - 4d).$$

Fact A.13 gives $k \ge \frac{38}{23}$, $P > C$, $\delta \le d$, $U > 0$ and $V \ge 0$, with U, V as before. As in Lemma 7.3, s is not eligible. Since r_a is nonincreasing, $z_s \ge W - \delta$, so

$$\begin{aligned} G &:= k(4-k)\delta - p - (k-2)^2 d + (4-k)(3-k)d_0 \\ &= (4-k)\big[(k-1)(d+\delta) + (4-k)d_0 - z_s\big] + (4-k)(z_s - W + \delta) \;\ge\; 0, \end{aligned}$$

and Fact A.13 gives $Z := P - (k-1) + (k-2)^2(d+\delta) - (4-k)^2 d_0 > 0$. The rows now follow:

- *Eligible t:* $M \ge U > 0$.
- *Ineligible $t \le s$:* $M \ge V \ge 0$, as $k > \frac{3}{2}$.
- *Ineligible $t \ge s$:* using $u \le 1 - T$, $z \le \mathrm{cap}$, $P' \ge P$ and $T \ge d + \delta$, we get $M \ge M_2 \ge P' - (k-1) + (k-2)^2 T - (4-k)^2 d_0 \ge Z > 0$. □

8 Proof of Theorem 2.2

By Lemma 3.2 the profile may be assumed strict. We run a procedure in which every step either returns a certificate, by Lemma 5.1, 7.3 or 7.4, or passes to a relayed carrier with a larger index. It therefore stops. Every certificate it returns has a point with at most two nonzero coordinates and at most $m + 3$ pieces.

Root. The root $a = 0$ is a carrier with $\lambda_0 = D = 0$, strong credit and an access path e_0.

- If $q_0 = 0$, apply (V).
- Otherwise let $q = \max\{1, q_0\} < 3$. An attainer s is sufficient-eligible: by Lemma 6.2(a) if $q = q_0$, and because $S_s(1) = w_s \geq 0$ if $q = 1$. Let c be the last active sufficient-eligible row. If $q \leq \frac{3}{2}$ or $v_c \geq \theta(q)$, apply (S). As $D = 0$, its credit requirement is $v_c \geq \max\{0, \theta\}$, and $\theta \leq 0 \leq v_c$ when $q \leq \frac{3}{2}$.
- Otherwise $q = q_0 > \frac{3}{2}$. If $q \leq 2$ and the last actual-eligible c' has $v_{c'} \geq \theta$, apply (A). If not, let s be an attainer. It is sufficient-eligible, so $v_s \leq v_c < \theta$, and Lemma 6.3 ($\lambda_0 \geq D$) makes $[e_0, e_s]$ strictly safe. Then s is a relayed carrier with parent 0:
  - $g_s > \frac{3}{4}$, and $g_s < 1$ by strictness;
  - s has strong credit by Lemma 6.1;
  - (H3) holds by the failed test at c' (if $q \leq 2$) or at c (if $q > 2$), since v is monotone;
  - (H4) is vacuous.

Relayed carrier h with parent a. Let $q = q_h$.

(i) $q = 0$: apply (V).

(ii) $0 < q \leq 1$: if some active $t > h$ is actual-eligible, apply (L). Otherwise $g_h < \frac{1281}{1600}$ by Lemma 6.2(c). Moreover $k \geq \frac{4}{3}$: else (H4), (H3) at h and Lemma 6.4(a) would give $g_h > \frac{1281}{1600}$. Apply Lemma 7.3.

(iii) $q \geq \frac{4}{3}$: an attainer s is sufficient-eligible (Lemma 6.2(a)). Apply (A) if $q \leq 2$ and (S) if $q > 2$ (then $\max\{2D, \theta\} = \theta$) at the last eligible $c \geq s$. If its credit test fails, then $v_s \leq v_c < \theta$, Lemma 6.3 makes $[e_h, e_s]$ strictly safe, and s is a relayed carrier with parent h.

(iv) $1 < q < \frac{4}{3}$ and $g_h \geq \frac{33}{40}$: an attainer s is actual-eligible (Lemma 6.2(b)). Apply (A) at the last actual-eligible $c \geq s$, or else relay to s by Lemma 6.3.

(v) $1 < q < \frac{4}{3}$, $g_h < \frac{33}{40}$ and $\lambda_h \geq 3(1 - g_h)$: by Lemma 6.2(a),(d), an attainer s is sufficient-eligible with $v_s \geq \theta$. So (A) succeeds at the last actual-eligible $c \geq s$.

(vi) $1 < q < \frac{4}{3}$, $g_h < \frac{33}{40}$ and $\lambda_h < 3(1 - g_h)$: if some active $t > h$ is actual-eligible, let c be the last such row, and apply (A) if $v_c \geq \theta_h(q)$. If there is no such row, or $v_c < \theta_h(q)$, apply Lemma 7.4, whose hypothesis holds because v^h is nondecreasing.

A relay in (iii) or (iv) produces a relayed carrier:

- $g_s < 1$ by strictness, and $g_s > \frac{3}{4}$;
- s has strong credit by Lemma 6.1, and an access path π_h followed by $[e_h, e_s]$;
- (H1) holds because $1 \leq q < 3$, and (H3) by the failed test;
- (H4) holds because relays with $q < \frac{4}{3}$ occur only in (iv).

This proves Theorem 2.2, and with Proposition 2.3, Theorem 1.2. □

A Scalar facts

All variables are real. Each fact is followed by its certificate. Bernstein coefficients are taken on the stated interval, so positive coefficients prove positivity there.

Fact A.1 (Actual tail)**.** Let $1 \leq q \leq 2$, $\mu = q - \frac{q^2}{4}$, $\frac{3}{4} \leq A \leq 1$, $B = 4A - 3$, $\delta \geq 0$, $w \leq A - \delta$ and $\mu w < \mu A - B + (q - \mu)\delta$. Then $F := (3 - \mu)w - \mu\delta - (1 - \mu)A \leq \theta := 2A - 3B/q$.

Proof. With $Z = (q - \mu)\delta + \mu A - B - \mu w > 0$ and $Y = A - \delta - w \geq 0$: $q\mu(\theta - F) = \frac{1}{4}q^2(q - 1)(B - q\delta) + q(3 - \mu)Z$ (use if $q\delta \leq B$) and $q(\theta - F) = 3(q\delta - B) + q(3 - \mu)Y$ (use if $q\delta \geq B$). □

Fact A.2 (Exit)**.** Let $\frac{3}{4} \leq A \leq 1$, $\lambda, L \geq 0$, $0 \leq u \leq G \leq A$, $L + u \geq A + \lambda$, $0 \leq q < 2$, $\beta = 1 - \frac{q}{4}$ and $q(A - u) \geq 3G - 4u - L + 4A - 3$. Suppose $\lambda \geq \frac{3}{2}(1 - A)$, or $q \geq 1$ and $A + \lambda \geq 1$. Then $u(L + G) \geq G^2$ or $2\beta u \leq G$.

Proof. Suppose not. Then $G, u > 0$ and $r := u/G \in (\frac{1}{2\beta}, 1)$. Put

$$\Gamma = r + \tfrac{1}{r} - 1 \geq 1, \qquad D_r = 4 - \tfrac{1}{r} + 4\beta(1-r) > 0, \qquad \Psi = r + \Gamma - \tfrac{4-1/r}{4\beta}, \qquad J = L + u.$$

- From $G^2 - u(L+G) = u(G\Gamma - J)$: $J < G\Gamma$.
- The hypothesis is $A \leq u + \frac{3-3G+L}{4\beta}$; with $L < G\Gamma - u$ this gives $A < \bar{A} := Gr + \frac{3-G(4-1/r)}{4\beta}$.
- Then $G \leq A < \bar{A}$ gives $GD_r < 3$.
- As $\frac{1}{4\beta} < \frac{r}{2}$: $\Psi > \frac{1}{r} - \frac{1}{2} > 0$.

If $\lambda \geq \frac{3}{2}(1-A)$:

$$3 \leq 3A + 2\lambda \leq A + 2J < \bar{A} + 2G\Gamma = \tfrac{3}{4\beta} + G(\Psi + \Gamma) < \tfrac{3}{4\beta} + \tfrac{3(\Psi+\Gamma)}{D_r} = \tfrac{3(1+2\Gamma)}{D_r} < 3,$$

using $\frac{D_r}{4\beta} + \Psi = 1 + \Gamma$ and $D_r - 1 - 2\Gamma = (1-r)((4\beta+2)r - 3)/r > 0$.

If $q \geq 1$ (so $\beta \leq \frac{3}{4}$) and $A + \lambda \geq 1$: $1 \leq J < G\Gamma < 3\Gamma/D_r < 1$, using $D_r - 3\Gamma = (1-r)((4\beta+3)r - 4)/r > 0$. □

Fact A.3 (Credit transfer)**.** Let $0 \leq d < \frac{1}{4}$, $A = 1 - d$, $B = 1 - 4d$, and $v, e, w \geq 0$ with $r := A - e - w \geq 0$, $v + w < \frac{1}{2}$ and $A - ez - rz(1-z) \geq \frac{3}{4}$ on $[0,1]$. Then $d + e \leq (v+w)(1-v-w)$.

Proof. At $z = 1$, $A - e \geq \frac{3}{4}$, so $r - e = 2(A-e) - A - w > 0$. Hence $z^* = \frac{e+r}{2r} \in [0,1]$, and the value at z^* gives $Br \geq (e+r)^2$, that is, $Be \leq (A-w)(w-3d)$. Now

$$B\big[(v+w)(1-v-w) - d\big] - (A-w)(w-3d) = (v+d)^2 + (1-2v-2w)\big[v + 2d(1-v-w)\big] \geq 0. \ \square$$

Attainer data. Facts A.4–A.6 share these hypotheses. Let $\frac{3}{4} \leq A = 1 - d \leq 1$, $V \geq 0$, $L = V + d$, $D = 3d$, $B = 1 - 4d$, and $0 < q < 3$. Let $\delta \geq 0$, w and v satisfy $q(\delta + w) = 4w - v - D$ and $v \geq L + w$, and put $E = B - q\delta - (q - \frac{q^2}{4})(A - \delta - w)$. Then $4E = qv + (q-2)^2 A - (4-q)D$ and $(3-q)v = (4-q)(v-w) + q\delta + D \geq (4-q)L + D$, so

$$4(3-q)E \ \geq \ \Phi(q, V, d) := (3-q)(q-2)^2 + q(4-q)V + (q^3 - 11q^2 + 44q - 48)\, d. \tag{A.1}$$

Fact A.4 (Eligibility at $\frac{33}{40}$)**.** With attainer data, $0 < d \leq \frac{7}{40}$, $V \geq d$ and $1 \leq q \leq 2$: $E > 0$.

Proof. As $V \geq d$, $\Phi \geq (3-q)(q-2)^2 + \kappa(q)d$, where $\kappa(q) = q^3 - 12q^2 + 48q - 48$. This κ is increasing ($\kappa' = 3(q-4)^2$) and $\kappa(\frac{3}{2}) = \frac{3}{8}$.

- If $\kappa(q) \geq 0$: $\Phi \geq (3-q)(q-2)^2 > 0$ for $q < 2$, and $\Phi \geq 8d > 0$ at $q = 2$.
- If $\kappa(q) < 0$: then $q < \frac{3}{2}$ and $\Phi \geq (3-q)(q-2)^2 + \frac{7}{40}\kappa(q) = \frac{1}{40}(144 - 304q + 196q^2 - 33q^3)$. Its Bernstein coefficients on $[1, \frac{3}{2}]$ are $\frac{3}{40}, \frac{7}{240}, \frac{89}{480}, \frac{141}{320}$. □

Fact A.5 (Eligibility at $\frac{1281}{1600}$)**.** With attainer data, $A \geq \frac{1281}{1600}$, $0 < q \leq 1$, and $V \geq \frac{1}{2}$ or $d \leq V(1-V)$: $E > 0$.

Proof. Let $V_0 = \frac{1}{2}(1 - \sqrt{1-4d})$. Then $V \geq V_0$, and $d \leq \frac{319}{1600} = \frac{11}{40} \cdot \frac{29}{40}$ gives $V_0 \leq \frac{11}{40}$. Since Φ increases in V and $d = V_0(1-V_0)$, $\Phi \geq \varphi(V_0)$ with $\varphi(z) := \Phi(q, z, z(1-z))$. On $[0, \frac{11}{40}]$, $\varphi'(z) = q(4-q) + (q^3 - 11q^2 + 44q - 48)(1-2z) \leq 3 - 14 \cdot \frac{9}{20} < 0$, because $3 - q(4-q) = (1-q)(3-q) \geq 0$ and $q^3 - 11q^2 + 44q - 48 + 14 = (q-1)(q^2 - 10q + 34) \leq 0$. So $\Phi \geq \varphi(\frac{11}{40}) = \frac{1}{1600}(3888 - 9804q + 7251q^2 - 1281q^3)$. Its Bernstein coefficients are all at least $\frac{1}{3200}$:

$$[0, \tfrac{1}{2}]: \ \tfrac{243}{100}, \tfrac{1127}{800}, \tfrac{4897}{6400}, \tfrac{5109}{12800}; \qquad [\tfrac{1}{2}, \tfrac{3}{4}]: \ \tfrac{5109}{12800}, \tfrac{5533}{25600}, \tfrac{5249}{51200}, \tfrac{4689}{102400};$$
$$[\tfrac{3}{4}, \tfrac{7}{8}]: \ \tfrac{4689}{102400}, \tfrac{3569}{204800}, \tfrac{269}{81920}, \tfrac{1473}{819200}; \qquad [\tfrac{7}{8}, 1]: \ \tfrac{1473}{819200}, \tfrac{1}{3200}, \tfrac{147}{12800}, \tfrac{27}{800}.$$ □

Fact A.6 (Baseline stop)**.** With attainer data, $1 \leq q \leq \frac{3}{2}$ and $L \geq D$: $v \geq 2A - 3B/q$.

Proof. $(3-q)v = (4-q)(v-w) + q\delta + D \geq (4-q)L + D \geq (5-q)D$, and $q(5-q)D - (3-q)(2Aq-3B) = A(3-2q)(3-q) + D(8q-q^2-9)$. If $8q-q^2-9 < 0$, rewrite the right side via $(3-2q)(3-q) + (8q-q^2-9) = q(q-1)$ as $Aq(q-1) + (A-D)(q^2-8q+9) \geq 0$. □

In Facts A.7–A.10, $\mathcal{E}(\beta,\omega) = \beta^2 - \beta(2A+2\omega-D) + D(A+\omega)$ with $A = 1-d$, $D = 3d$, $B = 1-4d$ and $L = V+d$.

Fact A.7 (Relay, $q \geq \frac{3}{2}$)**.** If $0 < V < \frac{1}{2}$ and $0 < d \leq V(1-V)$, then $\mathcal{E}(L, \frac{2}{3}(L+D)) < 0$.

Proof. $3\mathcal{E}(L, \frac{2}{3}(L+D)) = -V^2 - 6V + 7Vd + 3d + 17d^2$ increases in d. At $d = V(1-V)$ it equals $V(17V^3 - 41V^2 + 20V - 3)$. The cubic has Bernstein coefficients $-3, -\frac{4}{3}, -\frac{25}{48}, -\frac{19}{64}$ on $[0, \frac{1}{4}]$ and $-\frac{19}{64}, -\frac{7}{96}, -\frac{7}{16}, -\frac{9}{8}$ on $[\frac{1}{4}, \frac{1}{2}]$. □

Fact A.8 (Relay, $A \geq \frac{33}{40}$)**.** Let $0 < d \leq \frac{7}{40}$, $V > 0$, $L < D$ and $d \leq V(1-V)$. Let $1 \leq q < 3$, $b \geq L$, $\delta \geq 0$ and w satisfy $(3-q)w = q\delta + b + D$ and $(2A-b-w)q > 3B$. Then $w > D$ and $\mathcal{E}(L,D) < 0$.

Proof. $\mathcal{E}(L,D) = V^2 + Vd - 2V + d + 6d^2$ increases in d.

- If $V \leq \frac{9}{40}$: $d \leq V(1-V)$ gives $\mathcal{E}(L,D) \leq V(6V^3 - 13V^2 + 7V - 1)$. The cubic increases on $[0, \frac{9}{40}]$ and equals $-\frac{473}{32000}$ there.
- If $V > \frac{9}{40}$: then $V < 2d \leq \frac{7}{20}$, and $d \leq \frac{7}{40}$ gives $\mathcal{E}(L,D) \leq V^2 - \frac{73}{40}V + \frac{287}{800}$. This is convex, with values $-\frac{1}{800}$ and $-\frac{63}{400}$ at $\frac{9}{40}$ and $\frac{7}{20}$.

Suppose $w \leq D$. Then $w > 0$ (as $q < 3$ and $b + D > 0$), $b \leq 2w - D \leq D$, $q \leq 3 - \frac{b+D}{w}$ and $2A - b - w > 0$. Put $f_\beta(w) = (2A - \beta - w)(3 - \frac{\beta+D}{w})$. Then

$$3B < (2A-b-w)q \leq f_b(w) \leq f_L(w) \leq f_L(D) = 3B + \mathcal{E}(L,D)/D < 3B,$$

a contradiction. Here:

- $w(f_L(w) - f_b(w)) = (b-L)(2w + 2A - D - b - L) \geq 0$, since $2A - D - b - L > 2 - 11d > 0$;
- $Dw(f_L(D) - f_L(w)) = (D-w)[(2A-L)(L+D) - 3Dw] \geq 0$, since $(2A-L)(L+D) - 3D^2 = d(10-47d) + (L-2d)(2-7d-L) > 0$. □

Fact A.9 (Relay, $\frac{4}{3} \leq q \leq 2$)**.** Let $0 < V < \frac{1}{2}$, $\frac{V}{2} < d \leq V(1-V)$, $\frac{4}{3} \leq q \leq 2$ and $\theta = 2A - 3B/q$. Let $b \geq L$, $\delta \geq 0$ and w satisfy $(3-q)w = q\delta + b + D$ and $b + w < \theta$, and put $r = A - \delta - w$. Then $\frac{B}{4} - \delta z - rz(1-z) > 0$ for all $z \in [0,1]$.

Proof. Fix z. For fixed b the hypotheses describe the segment of the line $(3-q)w = q\delta + b + D$ with $w_\ell := \frac{b+D}{3-q} \leq w < w_\tau := \theta - b$, and the target is affine along it.

- At $w = w_\ell$ (where $\delta = 0$) the target is at least $\frac{1}{4}(B - \max\{r, 0\}) > 0$, as $r = A - w_\ell < A - D = B$. This uses $w_\ell \geq \frac{L+D}{3-q} > D$, from $L > 2d \geq (2-q)D$.
- At $w = w_\tau$ the target is affine in $b \in [L, b_{\max}]$, where $w_\ell(b_{\max}) = w_\tau(b_{\max})$; at $b_{\max}$ it is positive by the first point.

So it suffices to treat $b = L$, i.e. to show $J^*(q,z) \geq 0$, where

$$J^*(q,z) := \tfrac{B}{4} - \delta^* z - o^* z(1-z), \qquad w^* = \theta - L, \qquad \delta^* = \tfrac{(3-q)w^* - L - D}{q}, \qquad o^* = A - \delta^* - w^*.$$

At $q = \frac{4}{3}$: $w^* = -\frac{1}{4} + 6d - V$, $\delta^* = -\frac{5}{16} + \frac{9}{2}d - 2V$, $A - w^* = \frac{5}{4} - 7d + V$ and $o^* - \delta^* = \frac{15}{8} - 16d + 5V$. Also $J^*(0) = \frac{B}{4} > 0$ and $J^*(1) = \frac{1}{16}(9 - 88d + 32V) \geq \frac{1}{16}(88(V - \frac{7}{22})^2 + \frac{1}{11}) > 0$. A negative interior minimum needs $o^* > 0$, $A - w^* = \delta^* + o^* > 0$, $o^* - \delta^* > 0$ and $M^* := Bo^* - (A - w^*)^2 < 0$. Now $M^* = -V^2 + 2dV + \frac{V}{2} - \frac{d}{4} - 3d^2$ is concave in d, and:

- at $d = \frac{V}{2}$: $M^* = \frac{3}{8}V(1-2V) \geq 0$;
- at $d = V(1-V)$: $M^* = -\frac{1}{4}V(2V-1)^2(3V-1)$, which is ≥ 0 if $V \leq \frac{1}{3}$.

If $V > \frac{1}{3}$, the conditions force $d < d_\ell := \frac{5}{28} + \frac{V}{7}$ and $d < d_r := \frac{15}{128} + \frac{5V}{16}$. For $V \le \frac{55}{152}$ (where $d_r \le d_\ell$), $M^*(d_r) = \frac{171}{256}(V - \frac{7}{24})(\frac{55}{152} - V) \ge 0$. For $V \ge \frac{55}{152}$, at $d = d_\ell$ one has $A - w^* = 0$, $B = \frac{2-4V}{7}$, $\delta^* = \frac{55-152V}{112} \le 0$ and $M^* = -B\delta^* \ge 0$. By concavity $M^* \ge 0$ wherever needed, so $J^*(\frac{4}{3}, \cdot) \ge 0$.

For $q \ge \frac{4}{3}$: the point $(b, w, \delta) = (L, w^*, \delta^*)$ satisfies $v = \theta$, so (6.1) gives $3M^*(q) + \mathcal{E}^*(q) = 0$. Here $M^*(q) = Bo^* - (A - w^*)^2$ and $\mathcal{E}^* = \mathcal{E}(L, w^*)$, and

$$q\mathcal{E}^* = 3q(V^2 + 4Vd - 2V + d) + 3(-d + 4d^2 + 2V - 8Vd), \qquad V^2 + 4Vd - 2V + d \le -V(2V-1)^2 \le 0.$$

- If $\frac{4}{3}\mathcal{E}^*(\frac{4}{3}) \le 0$: then $q\mathcal{E}^*(q) \le 0$ for $q \ge \frac{4}{3}$, so $M^* \ge 0$. Hence $J^* \ge 0$: if $o^* > 0$, its minimum is $M^*/(4o^*)$, and if $o^* \le 0$ then $M^* \ge 0$ forces $o^* = \delta^* = 0$.
- Otherwise $V > \frac{1}{3}$, by the case $q = \frac{4}{3}$. Also $d > d_1 := \frac{8V+1}{20}$: the function $\frac{4}{3}\mathcal{E}^*(\frac{4}{3})$ is convex in d, equal to $-\frac{3}{2}V(1-2V) \le 0$ at $d = \frac{V}{2}$ and to $\frac{2}{25}(34V^2 - 19V + 1) \le 0$ at $d = d_1$. Now $F(q, z) := q^2 J^*(q, z)$ is quadratic in q with leading coefficient $A - \frac{3}{4} + z(A - L) + Az^2 > 0$, and

$$3\partial_q F(\tfrac{4}{3}, z) = (2 - 8d) + (20d - 8V - 1)z + (31d + 12V - 10)z^2.$$

  Its constant and linear coefficients are positive, and its coefficient sum $43d + 4V - 9 > \frac{424V-137}{20} > 0$; hence it is positive on $[0, 1]$. So F increases on $[\frac{4}{3}, \infty)$, and $J^*(q, z) \ge (\frac{4}{3q})^2 J^*(\frac{4}{3}, z) \ge 0$. □

Fact A.10 (Small parent)**.** Let $\frac{33}{40} \le A < 1$, $V \ge \frac{1}{2}$ or $d \le V(1-V)$, $1 \le k \le \frac{4}{3}$, and let $e \ge 0$, W and $b \ge L$ satisfy $(3-k)W = ke + b + D$ and $b + W < \theta := 2A - 3B/k$. Then $A - e > \frac{27}{20} - \sqrt{3/10} > \frac{1281}{1600}$.

Proof. From $b + W < \theta$: $ke < (3-k)\theta - (4-k)L - D$, i.e. $e < e_*(k) := -(2A - L) + \frac{9A-4L-4D}{k} - \frac{9B}{k^2}$.

- e_* is increasing. Indeed $k^3 e_*' = 18B - k(9A - 4L - 4D)$, and $3(18B - k(9A - 4L - 4D)) \ge 54B - 4(9A - 4L - 4D) = 18A - 38D + 16L \ge 18 - 100d > 0$ when $9A - 4L - 4D \ge 0$ (as $L \ge 2d$).
- Hence $e < e_*(\frac{4}{3}) = -\frac{5}{16} + \frac{9}{2}d - 2V$, i.e. $A - e > \frac{21}{16} - \frac{11}{2}d + 2V$.

Let $u_0 = \frac{1}{2}(1 - \sqrt{3/10})$, so $u_0(1 - u_0) = \frac{7}{40}$.

- If $V \ge u_0$: $d \le \frac{7}{40}$ gives $A - e > \frac{7}{20} + 2u_0$.
- If $V < u_0$: $d \le V(1-V)$ gives $A - e > h(V) := \frac{21}{16} - \frac{7}{2}V + \frac{11}{2}V^2$, and $h(V) - h(u_0) = (u_0 - V)(\frac{7}{2} - \frac{11}{2}(V + u_0)) \ge 0$, with $h(u_0) = \frac{7}{20} + 2u_0$.

Finally $\frac{7}{20} + 2u_0 = \frac{27}{20} - \sqrt{3/10}$, and $(\frac{879}{1600})^2 - \frac{3}{10} = \frac{4641}{2560000} > 0$. □

Fact A.11 (Baseline upgrade)**.** Let $\frac{3}{4} < A \le 1$, $d_0 = 1 - A$, $D_0 = 3d_0$, $B_0 = 1 - 4d_0$, $\lambda \ge 2d_0$ and $1 \le k < 3$. Let $e \ge 0$ with $A - e \ge \frac{3}{4}$, and W, $b \ge \lambda$ with $(3-k)W = ke + b + D_0$, $b + W < 2A - 3B_0/k$ and $b + W + e < D_0 + 3e$. Then $\frac{4}{3} < k < 2$ and $d_0 < \frac{1}{8}$.

Proof. $(3-k)(b + W + e - D_0 - 3e) = (4-k)b - (2-k)(D_0 + 3e)$ and $D_0 + 3e \le \frac{3}{4}$. For $k \ge 2$ the right side is ≥ 0; for $k \le 2$ and $\lambda \ge \frac{1}{4}$ it is $\ge \frac{4-k}{4} - \frac{3}{4}(2-k) = \frac{k-1}{2} \ge 0$. So $k < 2$ and $\lambda < \frac{1}{4}$, hence $d_0 < \frac{1}{8}$.

If $k \le \frac{4}{3}$, then $(3-k)(b + W) = (4-k)b + ke + D_0 \ge (11 - 2k)d_0$, so

$$k(3-k)\big[b + W - 2A + \tfrac{3B_0}{k}\big] \ge (2 - 4d_0)k^2 + (29d_0 - 9)k + 9 - 36d_0 \ge \tfrac{1}{8}(3k - 4)(4k - 9) \ge 0.$$

The middle step uses that the d_0-coefficient $-4k^2 + 29k - 36$ is negative on $[1, \frac{4}{3}]$. This contradicts $b + W < 2A - 3B_0/k$. □

Fact A.12 (Low rollback)**.** Assume:

(i) $\frac{4}{3} \le k < 3$, $0 \le d_0 \le d$, $\frac{319}{1600} < d \le \frac{1}{4}$, and $V_a \ge 0$ with $V_a \ge \frac{1}{2}$ or $d_0 \le V_a(1 - V_a)$;

(ii) $p = (4-k)W - kd - (4-k)d_0$, $p \ge W + V_a$ and $p < C = 2 - \frac{3}{k} + \frac{3(4-k)d_0}{k}$;

(iii) $0 < q \le 1$, $\delta \ge 0$, $0 \le w \le 1 - d - \delta$, $P = (4-q)w - q\delta - 4d$, $P \ge p + w$ and $qP \le (4-q)^2 d - (q-2)^2$.

Then $\frac{40}{23} < k < \frac{40}{17}$, $P > C$, $U > 0$ and $Y > 0$, and $V > 0$ if $k \le 2$, $V' > 0$ if $k \ge 2$ (U, V, V', Y as in Lemma 7.3).

Proof. Sources. The following are nonnegative:

$$F_1 = (3-k)p - kd - 2(4-k)d_0 = (4-k)(p - W - d_0),$$
$$F_2 = (3-q)P - q\delta - 4d - (4-q)p = (4-q)(P - p - w),$$
$$F_3 = (4-q)(1-d) - 4\delta - 4d - P = (4-q)(1 - d - \delta - w),$$
$$F_4 = (4-q)^2 d - (q-2)^2 - qP.$$

Here $V_a \ge d_0$. Let $c = (3-q)(4-q)^2 - 4q = 14 + 25(1-q) + 8(1-q)^2 + (1-q)^3$, $n = (3-q)(q-2)^2$, $a = q(4-q) \le 3$ and $D_q = q^2 - 3q + 3 > 0$. Then

$$J := qF_2 + (3-q)F_4 = cd - n - ap - q^2\delta \ge 0,$$

and $F_5 := (5-3q)d - p - D_q\delta \ge 0$, by $4(4-q)F_5 = (q-2)^2[(3-q)F_3 + F_2] + (4-q)J$.

Range of k; sign of $\eta := 20 - 11k + (2k-5)q$. $F_1 \ge 0$ and $p < C$ give $(d+2)k^2 - 9k + 9 < (4-k)(9-5k)d_0$.

- For $k \ge \frac{9}{5}$ the right side is ≤ 0. The left side increases in d and equals $\frac{9}{1600}(23k-40)(17k-40)$ at $d = \frac{319}{1600}$, so $\frac{40}{23} < k < \frac{40}{17}$. For $k \ge \frac{20}{11}$, $\eta < 0$ is clear.
- Let $\frac{4}{3} \le k \le \frac{20}{11}$. Put $R = (3-k)c - ka \ge \frac{122}{11}$, $\beta = (9-4k)c - 3ka \ge \frac{86}{11}$, $L' = \beta - kc$ and $\Sigma = k^2 n - (2k-3)R$. The sources $F_{\rm par} = (3-k)p - kd - (4-k)(d_0 + V_a)$, $F_{\rm cr} = k(C-p)$ and $cd - n - ap = J + q^2\delta$ are nonnegative, and

  $$kc\,F_{\rm par} + k^2(cd - n - ap) + R\,F_{\rm cr} = -\Sigma - (4-k)(kcV_a - \beta d_0).$$

  With $v = \min\{V_a, \frac{1}{2}\}$, we have $d_0 \le v(1-v)$ and $V_a \ge v$, so $\Sigma + (4-k)(\beta v^2 - L'v) \le 0$ $(*)$. Σ is convex in k: with $x = 1-q$, $\partial_k^2\Sigma = 6(12+17x+6x^2+x^3)$ and $-\partial_k\Sigma(\frac{20}{11}, q) = \frac{3}{11}(15+123x+13x^2-7x^3) > 0$, so Σ decreases in k. If $\eta \ge 0$, then $k \le K_\eta := \frac{5(4-q)}{11-2q} \le \frac{20}{11}$ and $\Sigma \ge \Sigma(K_\eta) = \frac{3P_\eta(q)}{(11-2q)^2}$, with $P_\eta = 144 - 468q + 637q^2 - 363q^3 + 84q^4 - 7q^5$; the Bernstein coefficients of $10P_\eta$ are $1440, 504, 205, 180, 234, 270$. If $k \le \frac{40}{23}$, then $\Sigma \ge \Sigma(\frac{40}{23}) = \frac{3}{529}(18 + 285y + 1136y^2 - 427y^3) > 0$, where $y = q - 1 \le 0$ and $285^2 < 4 \cdot 1136 \cdot 18$. In both cases $\Sigma > 0$.
  - If $L' \le 0$, this contradicts $(*)$.
  - If $L' > 0$, then $k < K_L := \frac{9c}{5c+3a} \le \frac{9}{5}$, and $4\beta[\Sigma + (4-k)(\beta v^2 - L'v)] = (4-k)(2\beta v - L')^2 + Z$ with $Z := 4\beta\Sigma - (4-k)L'^2 = k\,J_Z(k,q)$. Here J_Z is concave in k: $-\frac{1}{2}\partial_k^2 J_Z = 3119 + 7994x + 8025x^2 + 4444x^3 + 1473x^4 + 266x^5 + 23x^6$. The Bernstein coefficients of $135J_Z(\frac{4}{3}, q)$ are $34560, 6720, 15280, 33162, 46805, 50710, 43980$, and $J_Z(K_L, q) = \frac{36c^3(8+17y+14y^2-7y^3)}{(5c+3a)^2} > 0$. So $Z > 0$, again contradicting $(*)$.

  Hence $k > \frac{40}{23}$ and $\eta < 0$.

$P > C$. From $F_2 \ge 0$ and $P \ge p$, $3P \ge 4p + 4d$; with $F_1 \ge 0$ and $d \ge \frac{1}{6}$, $P - C \ge \frac{6k^2 - 25k + 27 + (4-k)(17k-27)d_0}{3k(3-k)} > 0$, as $6k^2 - 25k + 27 = 6(k - \frac{25}{12})^2 + \frac{23}{24}$ and $k > \frac{27}{17}$.

$U > 0$. For $k \le \frac{20}{11}$:

$$k(3-k)U = k(9-2k)(p - W - d_0) + k(k^2 - k + 3)d + (-k^3 - 4k^2 + 51k - 72)d_0 + (4k-6)(k-3).$$

Using $d > \frac{319}{1600}$ and $0 \le d_0 \le d$, the last three terms are at least $N_0 = \frac{1}{1600}(319k^3 + 6081k^2 - 27843k + 28800)$ if the d_0-coefficient is ≥ 0. Otherwise they are at least $N_1 = \frac{1}{1600}(4805k^2 - 11574k + 5832)$, and then the sum $-5k^2 + 54k - 72$ of the two coefficients is positive. Bernstein coefficients on $[\frac{40}{23}, \frac{20}{11}]$:

- $-5k^2 + 54k - 72$: $\frac{3592}{529}, \frac{2084}{253}, \frac{1168}{121}$;
- N_0: $\frac{136213}{486680}, \frac{101179}{465520}, \frac{4589}{27830}, \frac{1187}{9680}$;
- N_1: $\frac{15631}{105800}, \frac{27799}{101200}, \frac{5087}{12100}$.

For $k \ge \frac{20}{11}$: $p \ge \frac{kd}{3-k} > \frac{3}{10}$, and $p < C \le \frac{5}{4}$ since $\frac{5}{4} - C = \frac{3(4-k)(1/4-d_0)}{k}$. Write $cd - n - ap = c\,F_6 + \mathcal{B}(q,p)$, with $F_6 := d - \frac{3}{16} - \frac{p}{20}$ and $\mathcal{B}$ affine in p. Then $\mathcal{B}(q, \frac{3}{10}) = -\frac{1}{400}(26 + 265y + 832y^2 - 319y^3) < 0$,

since $265^2 < 4 \cdot 832 \cdot 26$, and $\mathcal{B}(q, \frac{5}{4}) = -\frac{3}{4}q^2(4-q) \le 0$. So $F_6 \ge 0$, and

$$U = \frac{(k^2-44k+184)F_1}{3(4-k)(20-7k)} + \frac{20(k^2-k+3)F_6}{3(20-7k)} + \frac{(-19k^3-91k^2+1052k-1440)d_0}{3k(20-7k)} + \frac{5k^3+107k^2-473k+480}{4k(20-7k)}.$$

All coefficients are positive on $[\frac{20}{11}, \frac{5}{2}]$:

- $k^2-44k+184 \ge \frac{321}{4}$;
- the d_0-cubic is concave, with end values $\frac{76800}{1331}$ and $\frac{2595}{8}$;
- $5k^3+107k^2-473k+480 = 5(k-\frac{20}{11})y'^2 + \frac{1497}{11}y'^2 + 15y' + 2$ with $y' = k-2$.

V, V'. For $k \le 2$ (so $k > \frac{3}{2}$ and $\eta < 0$), put $B' = 2k-3-(2-k)D_q$, $A' = (3-k)D_q-(2k-3)(5-3q)$ and $F' = (3-k)A' + kB'$. Then

$$(3-k)D_qV = [(3-k)D_q + B']F_1 + F'd + 2(4-k)B'd_0 + (3-k)(2k-3)F_5,$$
$$(11-2q)B' = -(D_q+2)\eta + (1-q)(1+q^2) > 0,$$
$$8(q-3)^2F' = [4(q-3)^2(k-2)+3q]^2 + q^2[8(q-3)^2-9] > 0.$$

For $k \ge 2$, put $x = k-2$ and $L_x = (1-q)-(2-q)x$. Then $(1-x)F_5+F_1 = 3L_xd-(1-x)D_q\delta-2(2-x)d_0$ shows $L_x \ge 0$, so $x \le \frac{1}{2}$ and $q \le \frac{1-2x}{1-x}$. With $E_x = (1-x)q^2+(6x-9)q+12(1-x)$, which is positive because $(2-q)E_x = (q^2-6q+12)L_x + 4D_q$,

$$nE_xV' = n(4-q)(4-q-x^2)F_1 + N_\beta F_5 + (N_\rho + xE_x)J + n(4-k)N_\eta d_0 + nN_\rho,$$

where

- $N_\rho = q^2(1+x) + (2-x)x^2(q-3)^2 > 0$;
- $N_\eta = 8+5x+20x(\frac{1}{2}-x) + q^2x(3-x) + 8qx^2 + 2q + 21x(1-q) > 0$;
- $N_\beta = (48-q^3)(1-x)^2(1+x) + q^2(11x^3-14x^2-12x+12) - 4q(1-x^2)(13-10x) > 0$. For the last: $\partial_q N_\beta \le -18x^3+24x^2+16x-28 < 0$, and at $q = \frac{1-2x}{1-x}$, $N_\beta = \frac{7-2x-5x^2+x^3-8x^4+4x^5}{(1-x)^2} > 0$.

$Y > 0$. Let $x = k-2 \in (-\frac{3}{8}, \frac{3}{8})$, $\Delta_k = 3q^2-21q+36-k(q^2-6q+12) > 0$, R as above ($R \ge 2$), and

$$N_1 = (x-1)^2(x+1)q^2 + (-x^4-7x^3+13x^2+11x-7)q + 4x^4+16x^3-24x^2-16x+20,$$
$$N_3 = (x-1)^2(x+1)q - 3x^3+6x^2-12,$$
$$N_0 = (q^2-8q+20)k^2 + (-7q^2+53q-116)k + 10q^2-76q+144,$$
$$R_0 = (3-q)^2x^2(2-x) + q^2(1+x), \qquad G = qN_1 - (k-1)^2(3-k)(4-q)\Delta_k,$$
$$R_d = -(3-q)(2-q)^2x^4 + 3(q-2)^2x^3 + (2-q)(5q^2-25q+54)x^2 - 3(q^3-3q^2-4q+4)x - 2q^3+14q^2-20q+12,$$

and $A_F = 4k(4-q)^2(4-q-x^2)$, $A_E = -k(4-q)N_3$, $A_d = -4k(4-k)(4-q)N_0$, $A_0 = 4k(4-q)R_0$, $h_E = (4-q)(kq-6k-3q+21)$. Then

$$4(4-q)\Delta_k\, kY = A_FF_1 + 4N_1F_2 - GF_3 + A_EF_4 + A_dd_0 + A_0,$$
$$4(4-q)\Delta_kR\, kY = (RA_F + 16(4-q)G)F_1 + (4RN_1 + 16(3-k)G)F_2 + 4(4-q)\Delta_kG\delta + (RA_E + h_EG)F_4 + (RA_d + 32(4-k)(4-q)G)d_0 + 4(4-q)\Delta_kR_d.$$

The signs on the domain are as follows.

- $N_1 \ge \frac{53}{8}$: the q-coefficient of N_1 is at most $-\frac{347}{512}$, so $N_1 \ge 3x^4+9x^3-11x^2-5x+13$.
- $N_3 \le -2x^3+5x^2-x-11 \le -9$.

- $N_0 < 0$: N_0 is convex in k with negative slope at $k = \frac{40}{17}$, so it decreases on $(K_\eta, \frac{40}{17}]$; and $N_0(K_\eta) = \frac{-(q-4)(5q^3-31q^2-19q-24)}{(11-2q)^2} < 0$.
- $R_0 > 0$ and $h_E \geq (4-q)(18-5k) > 0$.
- $R_d > 0$: its coefficients of $x^4, x^3, x^2, x, 1$ lie in $[-12, 0]$, $[0, 12]$, $[34, \infty)$, $[-12, 6]$ and $[\frac{20}{7}, \infty)$. Hence $R_d \geq 31x^2 - 12x + \frac{20}{7}$ for $x \geq 0$, and $R_d \geq 25x^2 + 6x + \frac{20}{7}$ for $x \leq 0$, both positive.

The first identity gives $Y > 0$ when $G \leq 0$, the second when $G \geq 0$. □

Fact A.13 (Middle rollback)**.** Assume:

(i) $\frac{4}{3} \leq k \leq 2$, $0 \leq d_0 \leq \frac{1}{8}$ and $\frac{7}{40} \leq d \leq \frac{1}{4}$;

(ii) $p = (4-k)W - kd - (4-k)d_0$, $p \geq W + d_0$, $W \geq d + 2d_0$, $p \leq 2d$ and $p \leq C$;

(iii) $1 \leq q \leq \frac{4}{3}$, $\delta \geq 0$, $0 \leq w \leq 1 - d - \delta$, $P = (4-q)w - q\delta - 4d$, $P \geq p + w$ and $q(2 - 3d - P) \geq 3(1 - 4d)$.

Then $k \geq \frac{38}{23}$, $P > C$, $\delta \leq d$, $U > 0$ and $V \geq 0$. If moreover $G := k(4-k)\delta - p - (k-2)^2 d + (4-k)(3-k)d_0 \geq 0$, then $Z := P - (k-1) + (k-2)^2(d+\delta) - (4-k)^2 d_0 > 0$.

Proof. Sources. F_1 and F_2 are as in Fact A.12, $F_3' = (3-q)(1-d) - 4d - p - 3\delta \geq 0$, and $J_m := qF_2 + (3-q)[q(2-3d-P) - 3(1-4d)] = (36 - 25q + 3q^2)d - (3-q)(3-2q) - q(4-q)p - q^2\delta \geq 0$. Also $p \geq W + d_0 \geq d + 3d_0$.

$k \geq \frac{38}{23}$. For $k < \frac{9}{5}$:

$$0 \leq C - p + \tfrac{3F_1}{2k} = 2 - \tfrac{3}{k} + \tfrac{(9-5k)p}{2k} - \tfrac{3d}{2} \leq 2 - \tfrac{3}{k} + (\tfrac{9}{k} - \tfrac{13}{2})d.$$

For $k \leq \frac{18}{13}$ the right side is at most $\frac{3}{8} - \frac{3}{4k} < 0$, using $d \leq \frac{1}{4}$. For $k > \frac{18}{13}$ it is at most $\frac{69}{80} - \frac{57}{40k}$, using $d \geq \frac{7}{40}$, which is negative for $k < \frac{38}{23}$.

$P > C$, $\delta \leq d$, $V \geq 0$. $2P = 3p + 4d + F_2 + q\delta + (q-1)(P-p)$ gives $P \geq \frac{7}{2}d + \frac{9}{2}d_0 \geq \frac{49}{80} + \frac{9}{2}d_0$, while $C \leq \frac{1}{2} + 3d_0$ (C increases in k); so $P - C \geq \frac{9}{80} > 0$. Next $(3-2q)F_3' + J_m = (3-q)[(5-q)d - (3-q)\delta - (q+1)p]$ and $p \geq d$ give $\delta \leq \frac{2(2-q)}{3-q}d \leq d$. Finally $V = F_1 + (2-k)(3d-p) + (2k-3)(d-\delta) \geq 0$.

Floors. As $36 - 25q + 3q^2 = 8 + 17(\frac{4}{3} - q) + 3(\frac{4}{3} - q)^2 > 0$, $J_m \geq 0$ gives $L_1 := d - \frac{2}{25} - \frac{2p}{5} \geq 0$ for $\frac{19}{80} \leq p \leq \frac{3}{10}$ and $L_2 := d - \frac{1}{8} - \frac{p}{4} \geq 0$ for $\frac{3}{10} \leq p \leq \frac{1}{2}$. Indeed, for the candidate floor $\ell(p)$, the quantity $(36 - 25q + 3q^2)\ell(p) - (3-q)(3-2q) - q(4-q)p$ is affine in p, with endpoint values $-\frac{3}{80}(3q-4)(11q-18)$, $-\frac{1}{10}(11q^2 - 28q + 18)$ and $-\frac{3}{4}q(q-1)$, all ≤ 0.

$U > 0$. For $\frac{38}{23} \leq k \leq \frac{7}{4}$:

$$\begin{aligned} 2k(4-k)U = {} & (24 - 3k - k^2)F_1 + (k^3 + 4k^2 - 51k + 72)(2d - p) \\ & + (-k^3 - 19k^2 + 134k - 144)(d - \tfrac{7}{40}) + \tfrac{1}{40}(-7k^3 + 187k^2 - 822k + 912). \end{aligned}$$

The coefficients are monotone on the interval, with minima $\frac{251}{16}$, $\frac{23}{64}$, $\frac{255720}{12167}$ and $\frac{555}{64}$.

For $\frac{7}{4} \leq k \leq 2$: here $p \geq \frac{kd}{3-k} \geq \frac{49}{200}$ and $p \leq 2d \leq \frac{1}{2}$, so $L_1 \geq 0$ or $L_2 \geq 0$, and

$$\begin{aligned} k(4-k)(15-7k)U = {} & k(2k^2 - 18k + 53)F_1 + 5k(4-k)(k^2 - k + 3)L_1 \\ & + (4-k)(-3k^3 - 42k^2 + 319k - 360)d_0 \\ & + \tfrac{2}{5}(4-k)(k^3 + 69k^2 - 252k + 225), \\ k(4-k)(12-5k)U = {} & k(k^2 - 12k + 40)F_1 + 4k(4-k)(k^2 - k + 3)L_2 \\ & + (4-k)(-3k^3 - 27k^2 + 236k - 288)d_0 \\ & + \tfrac{1}{2}(4-k)(k^3 + 39k^2 - 153k + 144). \end{aligned}$$

All coefficients are positive:

- the d_0-cubics are concave, with end values $\frac{3427}{64}, 86$ and $\frac{1679}{64}, 52$;
- with $x = k - \frac{7}{4}$, the constant cubics are $x^3 + \frac{297}{4}x^2 - \frac{21}{16}x + \frac{43}{64}$ and $x^3 + \frac{177}{4}x^2 - \frac{117}{16}x + \frac{67}{64}$, positive because their quadratic parts have negative discriminants.

$Z > 0$. $L_P := 2P - 3p - 4d - \delta = F_2 + (q-1)(P - p + \delta) \geq 0$, and with $\alpha = (k-2)^2$,

$$2k(4-k)Z = T + k(4-k)L_P + (1+2\alpha)G,$$
$$T := (13-\alpha)p + (16+5\alpha)d - (4-k)(27-k-2k^2)d_0 - 2k(4-k)(k-1).$$

For $\frac{4}{3} \leq k \leq \frac{7}{4}$, put $Q_0 = 27 - k - 2k^2$, $H_z = 2k^3 - 3k^2 - 38k + 63$, $D_z = -2k^3 + 9k^2 - 13k + 72$ and $E_z = (3-k)D_z - kH_z$. Then

$$2T = Q_0F_1 + H_z(2d-p) + (D_z - 2H_z)(d - \tfrac{7}{40}) + \tfrac{1}{40}N_+ \qquad (H_z \geq 0),$$
$$2(3-k)T = (3-k)Q_0F_1 - H_z[(3-k)p - kd] + E_z(d - \tfrac{7}{40}) + \tfrac{1}{40}N_- \qquad (H_z \leq 0),$$

where $(3-k)p - kd = F_1 + 2(4-k)d_0 \geq 0$ and

$$N_+ = 118k^3 - 695k^2 + 1081k - 378 = 7(D_z - 2H_z) - 160k(4-k)(k-1),$$
$$N_- = -160k^4 + 1196k^3 - 2494k^2 + 702k + 1512 = 7E_z - 160k(3-k)(4-k)(k-1).$$

Both decrease on $[\frac{4}{3}, \frac{7}{4}]$, to $\frac{567}{32}$ and $\frac{189}{16}$. So $N_\pm > 0$, hence also $D_z - 2H_z > 0$ and $E_z > 0$.

For $\frac{7}{4} \leq k \leq 2$, with L_1 or L_2:

$$\begin{aligned}(15-7k)T &= (5k^2 - 20k + 117)F_1 + 15(4-k)(2k^2 - 5k + 9)L_1 \\ &\quad + (4-k)(-14k^3 + 33k^2 + 164k - 171)d_0 + \tfrac{2}{5}(4-k)(35k^3 - 104k^2 + 60k + 27), \\ (12-5k)T &= (k^2 - 4k + 72)F_1 + 12(4-k)(2k^2 - 5k + 9)L_2 \\ &\quad + (4-k)(-10k^3 + 21k^2 + 139k - 180)d_0 + \tfrac{1}{2}(4-k)(20k^3 - 62k^2 + 33k + 27).\end{aligned}$$

All coefficients are positive on $[\frac{7}{4}, 2]$:

- the d_0-cubics are concave, with end values $\frac{4545}{32}, 177$ and $\frac{2367}{32}, 102$;
- $35k^3 - 104k^2 + 60k + 27$ is increasing from $\frac{69}{64}$;
- $20k^3 - 62k^2 + 33k + 27 = 20y^3 + 43y^2 - \frac{y}{4} + \frac{33}{16}$ with $y = k - \frac{7}{4}$. □

B Correspondence with the Lean development

Each statement is proved in the following modules of `ConsistentSubmodular/Deterministic/Certificates/`; the module `CompatibleProfiles` is in `ConsistentSubmodular/Deterministic/Basic/`.

Theorem 1.2	`WindowCertificates`
Definition 2.1, Theorem 2.2	`AtomicProfiles`, `FiniteCheckpoint`
Proposition 2.3	`CompatibleProfiles`, `CompletionReduction`, `ArrayCompletion`, `Triangularization`, `CompletionScaling`
Lemmas 3.1, 3.2	`AtomicFacts`, `Plateau`
Definition 4.1, Lemma 4.2	`Envelope`, `ActualConstructor`, `ActualCheckpoint`, `SuffConstructor`
Lemma 4.3	`TerminalSuff`, `TerminalActual`, `TerminalPairGeometry`
Lemma 5.1	`VertexCheckpoint`, `ConstructorTests`, `SuffScalar`
Lemmas 6.1–6.4	`Access`, `ProfileEligibility`, `RelayProfiles`, `SelectorRelays`
Definition 7.1, Lemma 7.2	`SelectorState`, `LowRollbackTargets`, `MiddleRows`, `RollbackCheckpoint`
Lemmas 7.3, 7.4	`LowRollback`, `MiddleRollback`
Section 8	`RootInitialization`, `SelectorLow`, `SelectorTermination`
Facts A.1, A.2, A.3	`ConstructorScalar`, `TerminalActualScalar`, `ScalarGeometry`
Facts A.4–A.6	`HighAnchor`, `EligibilityThreshold`, `BaselineStop`
Facts A.7–A.9	`RelayError`, `HighAnchor`, `FourThirdsBoundary`
Facts A.10, A.11	`SmallParent`, `BaselineUpgrade`
Fact A.12	`LowRollbackSources`, `LowRollbackExclusions`, `LowRollbackEligible`, `LowRollbackEligibleHigh`, `LowRollbackBefore`, `LowRollbackBeforeHigh`, `LowRollbackAfter`
Fact A.13	`MiddleSources`, `MiddleSigns`, `MiddleCredits`, `MiddleParent`